



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Indian Institute of Technology Jodhpur

CSL7490: Introduction to Blockchain

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Quiz 1

Time: 20 mins.

Date of Exam- 30th January, 2026

Maximum Marks: 20

Name and Roll no.:

- All questions are compulsory. Each question carries one mark.
 - Students must write their Name and Roll Number clearly on the question paper.
 - Each question must be answered in exactly one sentence only, written directly on the question paper.
 - No separate answer sheet shall be provided or accepted under any circumstances.
 - Wherever required, students must clearly state and justify any assumptions made in their answers.
 - Any form of copying or unfair means will result in zero marks in this quiz and may lead to further disciplinary action, including reduction of final course grades, as per the institute's rules.
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1. Which of the following statements best describes the relationship between distributed and decentralized systems?
 - a. All distributed systems are decentralized by default.
 - b. All decentralized systems are distributed, but not all distributed systems are decentralized.**
 - c. Distributed and decentralized are mutually exclusive terms.
 - d. A system must be centralized before it can become distributed or decentralized.
2. What is the main problem of a centralized system?
 - a. Trust & Privacy: Users must rely on the integrity of a single owner.
 - b. Single Point of Failure: If the central hub goes down, the entire system fails.**
 - c. Scalability: The central node limits the growth and speed of the entire network.
 - d. Censorship: The ability of a central authority to unilaterally control, suppress, or delete information and access within the network.
3. In a centralized social media platform, if a post is deleted by the company's moderators because it violates a new policy, this is an example of:
 - a. A Single Point of Failure
 - b. Scalability
 - c. Censorship**
 - d. Trustless Consensus
4. What is the Puzzle Friendly property of a PoW hash function designed to prevent?
 - a. It prevents miners from using analytical shortcuts to find a solution faster than brute force.**
 - b. It prevents nodes from communicating with each other during the mining process.
 - c. It prevents the network from growing too large.
 - d. All of the above

5. A system where every node performs identical computations to verify the same piece of data, but the final record is stored on a single master server, is best described as:
- Decentralized and Localized
 - Peer-to-Peer Distributed
 - Fully Decentralized
 - Distributed and Centralized**
6. In the Fair Exchange Problem, which scenario describes a failure to achieve the protocol's primary goal?
- Both parties decide to cancel the transaction before any data or assets are sent, resulting in a state where neither party loses or gains anything.
 - A network delay causes the exchange to take 24 hours to complete, even though both parties eventually receive their items.
 - An attacker intercepts a message and gains a copy of the digital item, but the original recipient still receives their copy as intended.
 - One party receives the intended asset, but then disconnects from the network before their own asset can be released to the other party.**
7. In the commitment-based Coin Flipping protocol, at which specific stage is the system most vulnerable to a Fairness Attack?
- Alice generates her random bit and salt but has not yet sent the hash to Bob.
 - Bob receives Alice's hash but refuses to send his own random bit in response.
 - Alice receives Bob's bit, calculates that the final result will be a "loss" for her, and refuses to reveal her original bit.**
 - Alice reveals her bit and salt, but Bob incorrectly claims that the hash does not match her original commitment.
8. The Avalanche Effect in a cryptographic hash function refers to which of the following behaviors?
- The hash value slowly changes as more data is added to the input.
 - The function becomes faster as the size of the input decreases.
 - Multiple inputs can be stacked to create a larger avalanche hash.
 - A small change in the input results in a significant and seemingly random change in the output.**
9. How does a Merkle Tree improve the efficiency of a blockchain node?
- It allows a node to verify if a specific transaction is included in a block without downloading the entire block data.**
 - It ensures that the block header remains in size regardless of the number of transactions.
 - It automatically corrects corrupted data within the block without needing to re-download it.
 - It increases the speed of the SHA-256 hashing algorithm through parallel processing.

10. What happens if a hash function used in a blockchain loses its Pre-image Resistance?

- a. The blockchain would grow faster because the blocks would become smaller.
- b. Miners would be able to solve Proof of Work puzzles instantly by working backward from the target hash.**
- c. Private keys would become public, allowing anyone to steal funds immediately.
- d. Transactions would be processed in parallel across different shards.

11. Why is the Collision Resistance of a hash function vital for digital signatures used in blockchain transactions?

- a. It ensures that the transaction ID is always exactly 32 bytes long.
- b. It makes the transaction process faster by reducing the amount of data the CPU has to read.
- c. It prevents the same transaction from being sent twice to the network.
- d. It prevents an attacker from creating two different transaction messages that result in the same digital fingerprint.**

12. Which cryptographic property ensures that a digital signature cannot be replayed or used for a different transaction by an attacker?

- a. The signature is only valid for exactly ten minutes.
- b. The signature uses a different private key for every individual node in the network.
- c. The signature is mathematically linked to the specific hash of the transaction data.**
- d. The signature is automatically deleted from the blockchain after it is verified.

13. What happens if a user's private key is stolen, even if they still have their public key and digital signature?

- a. The thief can generate valid signatures for new transactions, effectively gaining total control over the funds.**
- b. The user can use their public key to reset the private key through a consensus vote.
- c. The funds are safe because the digital signature is already on the blockchain.
- d. The network will detect the theft because the public key does not match the thief's IP address.

14. What is the primary benefit of using a Shared Distributed Ledger among multiple stakeholders (Drillers, Refiners, and Transporters) in the petroleum supply chain?

- a. It allows the refiner to remotely control the drill bits at the wellhead.
- b. It replaces the need for physical pipelines with digital assets.
- c. It physically speeds up the flow of oil through the pipelines.
- d. It provides a single, immutable Source of Truth for volume tracking, reducing reconciliation disputes.**

15. If a Public Trusted Bulletin Board loses its integrity property, what is the most likely consequence for a decentralized voting system?
- a. Voters will no longer be able to see who else has voted.
 - b. The system will become too slow to process a high volume of votes.
 - c. An attacker could secretly change a Yes vote to a No vote after it has been cast without being caught.
 - d. The electricity cost of running the voting servers will triple.

16. In a petroleum supply chain, replacing a single central authority with a peer-to-peer network to prevent any one company from unilaterally altering shipping records is known as _____.

Ans: Decentralization

17. In a decentralized network, the Integrity of a Public Trusted Bulletin Board is often enforced through a linked chain of hashes. This ensures that the board remains _____, meaning that new data can be added, but historical data cannot be edited or removed without invalidating the entire sequence.

Ans: Append-only or Immutable

18. The Blockchain Trilemma states that a single-layer network can only achieve two of the following three properties at once: _____, _____, and _____.

Ans: Decentralization

Security

Scalability

19. Traditional banking transfers through central hubs like the RBI/ NPCI face three primary issues: the risk of a _____, the burden of _____, and the inherent _____ for the participants.

Ans: Trusted Central Server or Trusted Third Party

High Transaction Fees

Lack of Anonymity

20. The Double Spending problem occurs when a user attempts to send the same digital token to two different recipients; blockchain solves this by using a _____ to maintain a single, immutable, and chronological record of all transactions.

Ans: Consensus Mechanism or Distributed Ledger